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SEMICONDUCTOR DEVICE AND METHOD OF ARRANGING AN INTERFACE OF A SEMICONDUCTOR DEVICE

Abstract

A semiconductor device including a first semiconductor die and a second semiconductor die is provided. The first semiconductor die includes a first substrate, a first interface circuit and first bonding components. The first bonding components are arranged in a first bond array region and the first interface circuit is arranged inside the first bond array region. The second semiconductor die includes a second substrate, a second interface circuit and second bonding components. The first bonding components are connected to the second bonding components, the first interface circuit and the second interface circuit are located between the first substrate and the second substrate, and the first semiconductor die and the second semiconductor die are interlinked by an interface constructed by the first interface circuit, the first bonding components, the second interface circuit and the second bonding components. A method of arranging an interface of a semiconductor device is also provided herein.

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Background/Summary

BACKGROUND

Technical Field

[0001] The disclosure is related to a semiconductor device and a method of arranging an interface of a semiconductor device.

Description of Related Art

[0002] With the development of the semiconductor industry, semiconductor devices enable more and more functions by the increase of integration. To achieve high integration, the interface for interlinking two or more semiconductor dies is required and the arrangement of the interface becomes one of the indispensable parts in IC (integrated circuit) design and manufacture.

SUMMARY

[0003] The disclosure is directed to a semiconductor device including an interface interlinking between two semiconductor dies with a flexible design on the arrangement of bonding components and interface circuits.

[0004] According to embodiments of the disclosure, a semiconductor device includes a first semiconductor die and a second semiconductor die. The first semiconductor die includes a first substrate, a first interface circuit and first bonding components, wherein the first bonding components are arranged in a first bond array region and the first interface circuit is arranged inside the first bond array region. The second semiconductor die includes a second substrate, a second interface circuit and second bonding components. The first bonding components are connected to the second bonding components, the first interface circuit and the second interface circuit are located between the first substrate and the second substrate, and the first semiconductor die and the second semiconductor die are interlinked by an interface constructed by the first interface circuit, the first bonding components, the second interface circuit and the second bonding components.

[0005] According to embodiments of the disclosure, the first interface circuit includes an interface logic circuit, transceivers and receivers, and the interface logic circuit, the transceivers and the receivers are arranged inside the first bond array region. The transceivers and the receivers are electrically connected to the first bonding components, respectively. A pitch of the transceivers and the receivers is smaller than a pitch of the first bonding components.

[0006] According to embodiments of the disclosure, a portion of the first interface circuit overlaps one or more of the first bonding components.

[0007] According to embodiments of the disclosure, the first semiconductor die further includes function circuit, the function circuit is in signal communication to the first interface circuit. A portion of the function circuit is arranged inside the first bond array region.

[0008] According to embodiments of the disclosure, the first semiconductor die further includes an interconnect wiring structure disposed between the first bonding components and the first interface circuit.

[0009] According to embodiments of the disclosure, the first bonding components and the second bonding components includes bonding pads, micro bumps or other types of 3D interconnection component.

[0010] According to embodiments of the disclosure, the second bonding components are arranged in a second bond array region substantially mirrored from the first bond array region. The second interface circuit is arranged inside the second bond array region. The second interface circuit includes an interface logic circuit, transceivers and receivers, and the interface logic circuit, the transceivers and the receivers are arranged inside the second bond array region. The transceivers and the receivers are electrically connected to the second bonding components, respectively. A pitch of the transceivers and the receivers is smaller than a pitch of the second bonding components.

[0011] According to embodiments of the disclosure, a portion of the second interface circuit overlaps one or more of the second bonding components.

[0012] According to embodiments of the disclosure, the second semiconductor die further includes a function circuit, and the function circuit is in signal communication to the second interface circuit. A portion of the function circuit is arranged inside a second bond array region of the second bonding components.

[0013] According to embodiments of the disclosure, the second semiconductor die further includes an interconnect wiring structure disposed between the second bonding components and the second interface circuit.

[0014] According to embodiments of the disclosure, a method of arranging an interface of a semiconductor device includes providing a first bond array region for arranging first bonding components implemented in a first semiconductor die; providing a first interface circuit implemented inside the first bond array region in the first semiconductor die; providing a second interface circuit and second bonding components implemented in a second semiconductor die; and bonding the first bonding components to the second bonding components.

[0015] According to embodiments of the disclosure, the first interface circuit includes an interface logic circuit, transceivers and receivers, and the interface logic circuit, the transceivers and the receivers are arranged inside the first bond array region. The transceivers and the receivers are electrically connected to the first bonding components, respectively, and a pitch of the transceivers and the receivers is smaller than a pitch of the first bonding components.

[0016] According to embodiments of the disclosure, the second bonding components are arranged in a second bond array region substantially mirrored from the first bond array region.

[0017] The second interface circuit includes an interface logic circuit, transceivers and receivers, and the interface logic circuit, the transceivers and the receivers are arranged inside the second bond array region. The transceivers and the receivers are electrically connected to the second bonding components, respectively, and a pitch of the transceivers and the receivers is smaller than a pitch of the second bonding components.

[0018] According to embodiments of the disclosure, a function circuit is further provided to be implemented inside the first bond array region in the first semiconductor die.

[0019] According to embodiments of the disclosure, a function circuit is further provided to be implemented in the second semiconductor die.

[0020] In light of the above, the semiconductor device includes semiconductor dies boded to each other in a face-to-face manner. The interface between the semiconductor dies include respective interface circuits and bonding components on each semiconductor die while the respective interface circuits are arranged inside the respective bond array regions of the bonding components.

Accordingly, the interface circuit is able to be compatible to various pitch designs of the bonding components and the substrate implemented with the interface circuit may have more flexibility on spacing design.

[0021] To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

[0023] FIG. 1 schematically illustrates a stack structure of a semiconductor device with the communication mechanism of the interface, according to an embodiment of the invention.

[0024] FIG. 2 schematically illustrates a communication mechanism of the interface between the first semiconductor die and an adjacent second semiconductor die, according to an embodiment of the invention.

[0025] FIG. 3 schematically illustrates an interface interlinking between the first semiconductor die and an adjacent second semiconductor die, according to an embodiment of the invention.

[0026] FIG. 4 schematically illustrates the arrangement of the interface for the semiconductor device in accordance with some embodiments of the disclosure.

[0027] FIG. 5 and FIG. 6 schematically illustrate the arrangements of the interface circuit and the bonding components in accordance with some embodiments of the disclosure.

[0028] FIG. 7 schematically illustrates a portion of a semiconductor die in a cross-section view in accordance with some embodiments of the disclosure.

[0029] FIG. 8 schematically illustrates a semiconductor device in a cross-sectional view in accordance with some embodiments of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

[0030] FIG. 1 schematically illustrates a stack structure of a semiconductor device with the communication mechanism of the interface, according to an embodiment of the invention.

Referring to FIG. 1, a semiconductor device **1000** includes a first semiconductor die **100** and multiple second semiconductor dies **200** and an interface **300** linking the first semiconductor die **100** to the second semiconductor dies **200**. The first semiconductor die **100**, such as a processor semiconductor die, is included as a base semiconductor die in the semiconductor device **1000**. The second semiconductor dies **200**, such as SRAM semiconductor dies, are stacked over the first semiconductor die **100**. The interface **300** is configured to communication with the information/data/signal between the first semiconductor die **100** and the second semiconductor dies **200**. In some embodiments, the interface **300** interlinking the first semiconductor die **100** to an adjacent second semiconductor die **200** may also be referred as Glink-3D which achieves the interlink between semiconductor dies in a 3D package structure.

[0031] In some embodiments, the first semiconductor die **100** of processor has a command to access the data stored the second semiconductor die **200** of SRAM. The first semiconductor die **100** may be considered as a master semiconductor die and the second semiconductor dies **200** may be considered as slave semiconductor dies, but the disclosure is not limited thereto. Due to the interface **300** as implemented, the read latency may be controlled to be about constant and small, such 2 ns or 5 ns in the examples. A single clock is used in the interface **300** to distribute to all the second semiconductor dies **200**, the path length from the first semiconductor die **100** to each second semiconductor die **200** is about the same and reliable. The latency can be adjusted to be about constant as predictable.

[0032] FIG. 2 schematically illustrates a communication mechanism of the interface between the first semiconductor die and an adjacent second semiconductor die, according to an embodiment of

the invention. Referring to FIG. 2, the first semiconductor die **100** may include a function circuit **110** and a first interface circuit **120** implemented therein. The function circuit **110** with the cache blocks **112** in an example form a processor and is in signal communication to the first interface circuit **120**. For example, the processor is connected to the first interface circuit **120** to transmit or receive signals through the first interface circuit **120**, as intended to communicate with the second semiconductor die **200**. The second semiconductor die **200** may include second function circuit **210** and a second interface circuit **220** implemented therein. The second function circuit **210** may include an SRAM circuit in signal communication to the second interface circuit **220** for communicating with the first semiconductor die **100**. In addition, the first semiconductor die **100** and the second semiconductor die **200** are connected through the bonding structure **400** which achieve the physical and electrical connection between the first semiconductor die **100** and the second semiconductor die **200**. In the embodiment, the first interface circuit **120**, the second interface circuit **220** and the bonding structure **400** construct the interface **300** which establishes a bi-way connection for signal transmission between the first semiconductor die **100** and the second semiconductor die **200**, and all signals are parallel transmitted or received by the interface **300**.

[0033] FIG. 3 schematically illustrates an interface interlinking between the first semiconductor die and an adjacent second semiconductor die, according to an embodiment of the invention. As shown in FIG. 3, the interface **300** includes the first interface circuit **120** implemented in the first semiconductor die **100** as indicated in FIG. 1 and the second interface circuit **220** implemented in the second semiconductor die **200** as indicated in FIG. 1 and the interface **300** further includes first bonding components **130** and second bonding components **230**. The first bonding components **130** are disposed in the first semiconductor die **100**, the second bonding components **230** are disposed in the second semiconductor die **200** and the first bonding components **130** are respectively bonded to the second bonding components **230** to serve as the bonding structure **400**.

[0034] In the embodiment, referring FIG. 1 to FIG. 3, the first semiconductor die **100** includes the function circuit **110**, the first interface circuit **120** and the first bonding components **130**. The function circuit **110** may include the cache circuit **112** therein to serve as a processor circuit as described in FIG. 2. The first interface circuit **120**, as shown in FIG. 3, includes an interface logic circuit **122**, transceivers **124** and receivers **126**. The transceivers **124** and the receivers **126** are connected to the first bonding components **130** in a one-to-one manner so as to establish the signal transmission to and/or from the second semiconductor die **200**. In some embodiments, the interface logic circuit **122** is configured to provide a control logic for BIST (built-in self-test), training, repair, parity functions and DFT (Design-for-Testability) logic, or the like.

[0035] In some embodiments, the first interface circuit **120** include flip-flop (FF) units, multiplexers and other components for establishing the interlink circuit to the second semiconductor die **200**. The multiplexer in an example is a double data rate (DDR) type in accordance with the input data at the flip-flop. As shown in FIG. 3, a single clock is provided through the first interface circuit **120** and the second interface circuit **220**. The flip-flop and the multiplexer are controlled in timing by the single clock.

[0036] In some embodiments, the first interface circuit **120** may receive a command as intended by the core circuit (e.g. the function circuit **110**) of the first semiconductor die **100** and provide outputs to the second semiconductor die **200** through the transceiver **124** and the corresponding bonding component **130**. The command in an example as an input may include a cluster of data Tx_data, a control signal such as IP control signal, and/or command without specific limitation. The command may also include a selecting slave identification, which is used to select the second semiconductor dies **200** to perform the command from the core circuit (e.g. the function circuit **110**) of the first semiconductor die **100**. The first interface circuit **120** may also receive the response from the second semiconductor die **200** through the corresponding bonding component **130** and the receiver **126**, and then inwardly transmit the data Rx_data into the core circuit (e.g. the function circuit **110**) of the first semiconductor die **100**.

[0037] In the embodiment, referring FIG. 1 to FIG. 3, the second semiconductor die **200** includes the function circuit **210**, the second interface circuit **220** and the second bonding components-**230**. The second function circuit **210** may include an SRAM circuit as described in FIG. 2. The second interface circuit **220**, as shown in FIG. 3, includes an interface logic circuit **222**, transceivers **224** and receivers **226**. The receivers **226** in the second interface circuit **220** are connected to the corresponding transceivers **124** in the first interface circuit **120** through the correspond bonding components **130** and the corresponding bonding components **230**. Similarly, the transceivers **224** in the second interface circuit **220** are connected to the corresponding receivers **126** in the first interface circuit **120** through the correspond bonding components **130** and the corresponding bonding components **230**. In addition, the second interface circuit **220** is controlled in timing by the single clock. As shown in FIG. 3, the interface logic circuit **222** may output the response, such as the data Rx_data to another second semiconductor die **220** (as shown in FIG. 1) and receive the data Tx_data from said another second semiconductor die **220**.

[0038] FIG. 4 schematically illustrates the arrangement of the interface for the semiconductor device in accordance with some embodiments of the disclosure. In FIG. 4, the semiconductor device **1000** includes the first semiconductor die **100** and the second semiconductor die **200**. Specifically, the first semiconductor die **100** and the second semiconductor die **200** are interlined by the interface **300**. The first semiconductor die **100** may include the function circuit **110**, the first interface circuit **120** and the first bonding components **130**. The second semiconductor die **200** may include the function circuit **210**, the second interface circuit **220** and the second bonding components **230**. FIG. 4 intends to present the arrangements of the components of the interface **300**, while the arrangements of the function circuit **110** and the function circuit **210** are not limited to the relationship shown in the drawings.

[0039] In the embodiment, the first bonding components **130** may be arranged in a prescribed pitch that is determined based on the bonding technique. Specifically, the first bonding components **130** are arranged in a first bond array region **R130**. The first interface circuit **120** is arranged in a region with a determined area smaller than the first bonding array region **R130** and is substantially completely located inside the first bonding array region **R130**. As described in above, the first bonding components **130** are connected to the transceivers **124** and the receivers **126** in the first interface circuit **120** in a one-to-one manner, and a pitch of the transceivers **124** and the receivers **126** may be smaller than a pitch of the first bonding components **130**. In some embodiments, a portion of the first interface circuit **120** overlaps one or more of the first bonding components **130**. In the embodiment, the function circuit **110** is in signal communication to the first interface circuit **120**. In some embodiments, a portion of the function circuit **110** may be arranged inside the first bonding array region **R130** and/or may overlap one or more of the first bonding components **130**.

[0040] The second bonding components **230** may be arranged in a prescribed pitch that is determined based on the bonding technique. Specifically, the second bonding components **230** are arranged in a second bond array region **R230**. The second bonding components **230** may be arranged corresponding to the first bonding components **130** so that the second bond array region **R230** is substantially mirrored from the first bond array region **R130**. Therefore, the second bonding components **230** are connected to the first bonding components **130** in a one-to-one manner. The second interface circuit **220** is arranged in a region with a smaller area than the second bonding array region **R230** and is substantially completely located inside the second bonding array region **R230**. As described in above, the second bonding components **230** are connected to the transceivers **224** and the receivers **226** in the second interface circuit **220** in a one-to-one manner. A pitch of the transceivers **224** and the receivers **226** may be smaller than a pitch of the second bonding components **230**. In some embodiments, a portion of the second interface circuit **220** overlaps one or more of the second bonding components **230**. In the embodiment, the function circuit **210** is in signal communication to the second interface circuit **220**. In some embodiments, the arrangement of the function circuit **210** may be arranged inside the second bonding array region

R230 and/or may overlap one or more of the second bonding components **230**.

[0041] In some embodiments, the arrangement region of the first interface circuit **120** may be independent from the arrangement of the first bonding component **130** and the arrangement region of the second interface circuit **220** may be independent from the arrangement of the second bonding component **230**. For example, FIG. 5 and FIG. 6 schematically illustrate the arrangements of the interface circuit and the bonding components in accordance with some embodiments of the disclosure. In FIG. 5, the interface circuit **20** may refer to the first interface circuit **120** or the second interface circuit **220** in the previous embodiment, the bonding components **30** may refer to the first bonding components **130** or the second bonding components **230** in the previous embodiments and the bonding array region **R1** may refer to the first bonding array region **R130** or the second bonding array region **230** in the previous embodiments. The bonding components **30** are arranged in a pitch **P1** in a 5×5 array and the bonding components **30** are arranged in the bonding array region **R1** that has an area **A1**. The interface circuit **20** is arranged in a prescribed region **R2** that has an area **A2**. As described in the previous embodiments, the interface circuit **20** may include an interface logic circuit, transceivers and receivers that may be fabricated using nano or smaller level semiconductor manufacturing process. Therefore, the physical implement of the interface circuit **20** requires a small space. As shown in FIG. 5, the area **A2** is smaller than the area **A1**. In the embodiment, the interface circuit **20** is completely arranged inside the bonding array region **R2**. In some embodiments, other circuits may be arranged at the space between the boundary of the area **A2** and the boundary of the area **A1** to optimize the spacing utility of the semiconductor device.

[0042] In another embodiment as shown in FIG. 6, the bonding components **30** are arranged in a pitch **P2** in a 5×5 array and the bonding components **30** are arranged in the bonding array region **R3** that has an area **A3**. Comparing FIG. 5 and FIG. 6, the pitch **P2** is larger than the pitch **P1** and the area **A3** is greater than the area **A2**, but the interface circuit **20** remains arranged in the prescribed region **R2** that has an area **A2**. It is noted that the interface circuit **20** may be applicable in various designs of the bonding array region without rearranging the components in the interface circuit **20**. In other words, the arrangement of the interface circuit **20** may be independent from the arrangement of the corresponding bonding components **30**. In some embodiments, other circuits may be arranged at the space between the boundary of the area **A2** and the boundary of the area **A3** to optimize the spacing utility of the semiconductor device.

[0043] FIG. 7 schematically illustrates a portion of a semiconductor die in a cross-section view in accordance with some embodiments of the disclosure. As shown in FIG. 7, a semiconductor die **500** includes a substrate **502**, electronic components for a function circuit **510**, electronic components for an interface circuit **520**, bonding components **530** and an interconnect wiring structure **504**. The substrate **502** may be a semiconductor substrate or a semiconductor substrate inclusive of isolation regions. The electronic components for the function circuit **510** and the electronic components for the interface circuit **520** are formed in the substrate **502**. The bonding components **530** are electrically connected to the electronic components in the interface circuit **520** through the interconnect wiring structure **504**.

[0044] The electronic components for the function circuit **510** and the electronic components for the interface circuit **520** may include active components such as transistors or the like, passive components such as capacitors, resistors or the like, or a combination thereof. In some embodiments, the electronic components for the function circuit **510** and the electronic components for the interface circuit **520** may be partially integrated in the substrate **502** and fabricated by using nano level or smaller semiconductor manufacturing process.

[0045] The interconnect wiring structure **504** include multiple metal layers **Mx** and multiple intervening dielectric layers to establish the required electrical transmission paths for the electronic components for the function circuit **510** and the electronic components for the interface circuit **520**. In some embodiments, the bonding components **530** include bonding pads that are optionally fabricated by the same method of forming the metal layers **Mx** in the interconnect wiring structure

504. In some embodiments, the bonding components **530** include micro bumps that is formed in a form of a conductor ball/pillar or the like.

[0046] In the embodiment, a portion of the electronic components of the interface circuit **520** is electrically connected to the corresponding bonding components **530** through the interconnect wiring structure **504** and such electronic components includes transceivers and receivers. In some embodiments, a portion of the electronic components of the interface circuit **520** overlaps one or more of the bonding components **530**. Specifically, the bonding components **530** are arranged in a bond array region **R530** and the interface circuit **520** is arranged inside the bond array region **R530**. The interface circuit **520** occupies a smaller area than the bond array region **R530** and the region between the outer most component of the interface circuit **520** and the boundary of the bond array region **R530** may be used by other chip level placement and routing to optimize the spacing utility of the substrate **502**.

[0047] The function circuit **510** may be disposed beside the interface circuit **520** and a portion of the function circuit **510** may be located inside the bond array region **R530**. In some embodiments, a portion of the electronic components of the function circuit **510** overlaps one or more of the bonding components **530**. The function circuit **510** may be in signal communication to the interface circuit **520** so that the electric signal from the function circuit **510** may be transmitted to the bonding component **530** through the interface circuit **520** and the signal input from the bonding component **530** may be transmitted to the function circuit **510** through the interface circuit **520**.

[0048] The interconnect wiring structure **504** includes multiple metal layers **Mx** that establish staggered signal transmission path between the electronic components of the interface circuit **520** and the bonding components **530**. The arrangement of the electronic components of the interface circuit **520** may be independent from the arrangement of the corresponding bonding components **530**. For example, when viewed from the top view, the electronic component (transceiver or receiver) of the interface circuit **520** and the corresponding bonding component **530** that are used for transmitting the same signal may be located at two separate positions and other components (transceiver or receiver) of the interface circuit **520** and/or bonding components **530** may be located between the two separate positions. In alternative embodiments, two adjacent electronic components (transceiver or receiver) of the interface circuit **520** may be spaced by a pitch that is smaller than the pitch of two corresponding bonding components **530** connecting thereto.

[0049] The structure of the semiconductor die **500** may be considered as an implemental example of the first semiconductor die **100** or the second semiconductor die **200** that are described in the previous embodiments. For example, the function circuit **510** may be an implemental example of the function circuit **110** or the function circuit **210** shown in FIGS. 2 to 4, the interface circuit **520** may be an implemental example of the first interface circuit **120** or the second interface circuit **220** shown in FIGS. 2 to 4, and the bonding components **530** may be an implemental example of the first bonding components **130** or the second bonding component **230** shown in FIG. 3. Accordingly, the interface circuit **520** and the bonding components **530** may be a part of the interface **300** depicted in FIGS. 1 to 4. In addition, the arrangement of the interface circuit **20** and the bonding components **30** depicted in FIGS. 5 and 6 may be applicable to the structure of the semiconductor die **500** so that the interface circuit **520** is compatible to various designs of bond array.

[0050] FIG. 8 schematically illustrates a semiconductor device in a cross-sectional view in accordance with some embodiments of the disclosure. A semiconductor device **2000** includes a first semiconductor die **500A** and a second semiconductor die **500B** that are bonded to each other in a face-to-face manner. Each of the first semiconductor die **500A** and the second semiconductor die **500B** may be similar to the semiconductor die **500** depicted in FIG. 7 and thus the descriptions for the components of the semiconductor die **500** in FIG. 7 are applicable to the corresponding components of the semiconductor device **2000**.

[0051] The first semiconductor die **500A** includes a first substrate **502A**, a first interface circuit **520A** and first bonding components **530A**, wherein the first bonding components **530A** are

arranged in a first bond array region **R530A** and the first interface circuit **520A** is arranged inside the first bond array region **R530A**. Specifically, the first semiconductor die **500A** further includes an interconnect wiring structure **504A** located between the first interface circuit **520A** and the first bonding components **530A**. The interconnect wiring structure **504A** includes metal layers establishing the signal transmission paths between the first interface circuit **520A** and the first bonding components **530A**.

[0052] The second semiconductor die **500B** includes a second substrate **502B**, a second interface circuit **520B** and second bonding components **530B**, wherein the second bonding components **530B** are arranged in a second bond array region **R530B** and the second interface circuit **520B** is arranged inside the second bond array region **R530B**. Specifically, the second semiconductor die **500B** further includes an interconnect wiring structure **504B** located between the second interface circuit **520B** and the second bonding components **530B**. The interconnect wiring structure **504B** includes metal layers establishing the signal transmission paths between the second interface circuit **520B** and the second bonding components **530B**.

[0053] In the embodiment, the first semiconductor die **500A** is oriented that the front side (the side implemented with circuits) faces up, and the second semiconductor die **500B** is oriented that the front side (the side implemented with circuits) faces down. The second bond array region **R530B** is substantially mirrored from the first bond array region **R530A**, and the first bonding components **530A** are connected to the second bonding components **530B** in a one-to-one manner. In the bonded structure as shown in FIG. 8, the first interface circuit **520A** and the second interface circuit **520B** are located between the first substrate **502A** and the second substrate **502B**, which is understood as a face-to-face bonded structure.

[0054] In the embodiment, the first bonding components **530A** and the second bonding components **530B** are bonding pads. The bonding face **FF** is formed by the first bonding components **530A** in contact with the second bonding components **530B** and the dielectric surrounding the first bonding components **530A** in contact with the dielectrics surrounding the second bonding components **530B**. In some alternative embodiments, the first bonding components **530A** and the second bonding components **530B** may be micro bumps or other types of 3D interconnection components and an underfill may be further disposed between the first semiconductor die **500A** and the second semiconductor die **500B** to surround the bonded structure formed by the first bonding components **530A** connecting to the second bonding components **530B**.

[0055] In the embodiment, the first semiconductor die **500A** and the second semiconductor die **500B** are interlinked by an interface **600** constructed by the first interface circuit **520A**, the first bonding components **530A**, the second interface circuit **520B** and the second bonding components **530B**. The interface **600** is considered as a 3D interlink technique for signal communication between semiconductor dies in a 3D stacking structure. The interface **600** may have a similar design to the interface **300** depicted in the previous embodiments.

[0056] In some embodiments, the first semiconductor die **500A** may be an implemental example of the first semiconductor die **100** shown in FIGS. 1-4. Referring to the previous embodiments, the first interface circuit **520A** may include the interface logic circuit **122**, the transceivers **124** and the receivers **126** shown in FIG. 3. The transceivers and the receivers in the interface circuit **520A** are electrically connected to the first bonding components **530A**, respectively. In addition, the first bonding components **530A** may be arranged in an array similar to the array shown in FIGS. 5 and 6, but the disclosure is not limited thereto. The interface logic circuit, the transceivers and the receivers in the interface circuit **520A** may be arranged inside the first bond array region **R530A**. In some embodiments, a pitch of the transceivers and the receivers in the interface circuit **520A** may be smaller than a pitch of the first bonding components **530A**. In some embodiments, a portion of the first interface circuit **520A** overlaps one or more of the first bonding components **530A**. In some embodiments, the arrangement of the transceivers and the receivers in the interface circuit **520A** may be independent from the arrangement of the first bonding components **530A**.

[0057] In some embodiments, the second semiconductor die **500B** may be an implemental example of the second semiconductor die **200** shown in FIGS. **1-4**. Referring to the previous embodiments, the second interface circuit **520B** may include the interface logic circuit **222**, the transceivers **224** and the receivers **226** shown in FIG. **3**. The second bonding components **530B** may be arranged in an array similar to the array shown in FIGS. **5** and **6**, but the disclosure is not limited thereto. The second interface circuit **520B** is arranged inside the second bond array region **R530B**. The interface logic circuit, the transceivers and the receivers in the second interface circuit **520B** are arranged inside the second bond array region **R530B**. The transceivers and the receivers in the second interface circuit **520B** are electrically connected to the second bonding components **530B**, respectively. A pitch of the transceivers and the receivers in the second interface circuit **520B** is smaller than a pitch of the second bonding components **530B**. A portion of the second interface circuit **520B** overlaps one or more of the second bonding components **530B**.

[0058] In some embodiments, a method of arranging the interface **600** of the semiconductor device **2000** may include providing a first bond array region **R530A** for arranging first bonding components **530A** implemented in a first semiconductor die **500A**. In some embodiments, the first bond array region **R530A** may be determined based on the prescribed pitch rule of arranging the first bonding components **530A**. For example, the first bond array region **R530A** is provided for arranging the first bonding components **530A** in a pitch ranged from 6 microns to 16 microns, but the disclosure is not limited thereto. In other alternative embodiments, the pitch of the first bonding components **530A** may be determined based on the design of the products.

[0059] Subsequently, the method further provides a first interface circuit **520A** implemented inside the first bond array region **R530A** in the first semiconductor die **500A**. The first interface circuit **520A** may include interface logic circuit, transceivers and receivers that are fabricated using the semiconductor manufacturing process and the electronic components in the first interface circuit **520A** may have small sizes at about nano level. Generally, the sizes and the connection relationships of the electronic components in the first interface circuit **520A** are determined in advance so that the area for arranging the interface circuit **520A** is determined.

[0060] In the embodiment, the first interface circuit **520A** is electrically to the corresponding first bonding components **530A** through the interconnect wiring structure **504A** shown in FIG. **8** and the interconnect wiring structure **504A** is able to provide detours for signal transmission. Therefore, the first interface circuit **520A** is arranged inside the first bond array region **R530A** without considering the relative positions of the first bonding components **530A**. In addition, the same specification on sizes and the connection relationships of the electronic components in the first interface circuit **520A** is compatible for various pitch designs of the first bonding components **530A**, which provides a flexible design rule for fabricating the first semiconductor die **500A** of the semiconductor device **2000**. Since the first interface circuit **520A** requires only a small area, the substrate may have more space for arranging other circuits or components, which improves the design flexibility of the semiconductor device **2000**.

[0061] The method also includes providing a second interface circuit **520B** and second bonding components **530B** implemented in a second semiconductor die **500B**. Similar to the first semiconductor die **500A**, the second bonding components **530B** in the second semiconductor die **500B** may be arranged in a second bond array region **R530B** and the second interface circuit **520B** in the second semiconductor die **500B** may be arranged inside the second bond array region **R530B**. In addition, the arrangement of the second interface circuit **520B** may be independent from the arrangement of the second bonding components **530B**. Therefore, the same specification on sizes and the connection relationships of the components in the second interface circuit **520B** is compatible for various pitch designs of the second bonding components **530B**. In some embodiments, the second bond array region **R530B** may be provided for arranging the second bonding components **530B** in a pitch ranged from 6 microns to 16 microns, but the disclosure is not limited thereto. In other alternative embodiments, the pitch of the second bonding components

530B may be determined based on the design of the products.

[0062] The method also includes bonding the first bonding components **530A** to the second bonding components **530B**. In the embodiments, the second bond array region **R530B** may be substantially mirrored from the first bond array region **R530A** and the first semiconductor die **500A** and the second semiconductor die **500B** are bonded in a face-to-face manner. As depicted in FIG. 8, the first semiconductor die **500A** further includes the interconnect wiring structure **504A** that establishes detour for signal transmission and thus the arrangement of the first interface circuit **520A** may be independent from the arrangement of the first bonding components **530A**. Similarly, the second interface circuit **520B** is electrically connected to the second bonding components **530B** through the interconnect wiring structure **504B** in the second semiconductor die **500B** and the arrangement of the second interface circuit **520B** may be independent from the second bonding components **530B**.

[0063] In view of the above, the semiconductor device in accordance with some embodiments of the disclosure includes multiple semiconductor dies bonded in a face-to-face manner. The interface interlinking between the semiconductor dies includes an interface circuit and bonding components formed in each of the semiconductor die. The interface circuit may be arranged in a determined region which is compatible to various pitch designs of the bonding components. Accordingly, the arrangement of the interface is flexible. In addition, the interface circuit is concentrated within the determined region so that the substrate implemented with the interface circuit may have flexible spacing utility.

[0064] It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. A semiconductor device, comprising: a first semiconductor die comprising a first substrate, a first interface circuit and first bonding components, wherein the first bonding components are arranged in a first bond array region and the first interface circuit is arranged inside the first bond array region; and a second semiconductor die comprising a second substrate, a second interface circuit and second bonding components, wherein the first bonding components are connected to the second bonding components, the first interface circuit and the second interface circuit are located between the first substrate and the second substrate, and the first semiconductor die and the second semiconductor die are interlinked by an interface constructed by the first interface circuit, the first bonding components, the second interface circuit and the second bonding components.
2. The semiconductor device of claim 1, wherein the first interface circuit comprises an interface logic circuit, transceivers and receivers, and the interface logic circuit, the transceivers and the receivers are arranged inside the first bond array region.
3. The semiconductor device of claim 2, wherein the transceivers and the receivers are electrically connected to the first bonding components, respectively.
4. The semiconductor device of claim 2, wherein a pitch of the transceivers and the receivers is smaller than a pitch of the first bonding components.
5. The semiconductor device of claim 1, wherein a portion of the first interface circuit overlaps one or more of the first bonding components.
6. The semiconductor device of claim 1, wherein the first semiconductor die further comprises function circuit, the function circuit is in signal communication to the first interface circuit.
7. The semiconductor device of claim 6, wherein a portion of the function circuit is arranged inside the first bond array region.
8. The semiconductor device of claim 1, wherein the first semiconductor die further comprises an

interconnect wiring structure disposed between the first bonding components and the first interface circuit.

9. The semiconductor device of claim 1, wherein the first bonding components and the second bonding components comprises bonding pads, micro bumps or other types of 3D interconnection components.

10. The semiconductor device of claim 1, wherein the second bonding components are arranged in a second bond array region substantially mirrored from the first bond array region.

11. The semiconductor device of claim 10, wherein the second interface circuit is arranged inside the second bond array region.

12. The semiconductor device of claim 10, wherein the second interface circuit comprises an interface logic circuit, transceivers and receivers, and the interface logic circuit, the transceivers and the receivers are arranged inside the second bond array region.

13. The semiconductor device of claim 12, wherein the transceivers and the receivers are electrically connected to the second bonding components, respectively.

14. The semiconductor device of claim 12, wherein a pitch of the transceivers and the receivers is smaller than a pitch of the second bonding components.

15. The semiconductor device of claim 1, wherein a portion of the second interface circuit overlaps one or more of the second bonding components.

16. The semiconductor device of claim 1, wherein the second semiconductor die further comprises a function circuit, and the function circuit is in signal communication to the second interface circuit.

17. The semiconductor device of claim 16, wherein a portion of the function circuit is arranged inside a second bond array region of the second bonding components.

18. The semiconductor device of claim 1, wherein the second semiconductor die further comprises an interconnect wiring structure disposed between the second bonding components and the second-interface circuit.

19. A method of arranging an interface of a semiconductor device, comprising: providing a first bond array region for arranging first bonding components implemented in a first semiconductor die; providing a first interface circuit implemented inside the first bond array region in the first semiconductor die; providing a second interface circuit and second bonding components implemented in a second semiconductor die; and bonding the first bonding components to the second bonding components.

20. The method of claim 19, wherein the first interface circuit comprises an interface logic circuit, transceivers and receivers, and the interface logic circuit, the transceivers and the receivers are arranged inside the first bond array region.

21. The method of claim 20, wherein the transceivers and the receivers are electrically connected to the first bonding components, respectively, and a pitch of the transceivers and the receivers is smaller than a pitch of the first bonding components.

22. The method of claim 19, wherein the second bonding components are arranged in a second bond array region substantially mirrored from the first bond array region.

23. The method of claim 22, wherein the second interface circuit comprises an interface logic circuit, transceivers and receivers, and the interface logic circuit, the transceivers and the receivers are arranged inside the second bond array region.

24. The method of claim 23, wherein the transceivers and the receivers are electrically connected to the second bonding components, respectively, and a pitch of the transceivers and the receivers is smaller than a pitch of the second bonding components.

25. The method of claim 19, further providing a function circuit to be implemented inside the first bond array region in the first semiconductor die.

26. The method of claim 19, further providing a function circuit implemented in the second semiconductor die.

